AIMS African Institute for Mathematical Sciences RWANDA

Introduction to Probability and Statistics

QUIZ # 3

2024-2025

Duration : 60 min.

Surname (in capital letters), first name		Score

Question 1 – Let X, Y and Z be three random variables and let a, b, c be three real numbers.

Express :

$E(aX - bY - cZ) =$

$Var(aX - bY - cZ) =$

Question 2 – Let X be a r.v. such that $X \in \left\{a, 2a, 3a\right\}$, $a > 0$, such that

$$P(X = a) = P(X = 2a) = P(X = 3a).$$

Calculate $E(X)$, the expectation of X :

Calculate $Var(X)$, the variance of X :

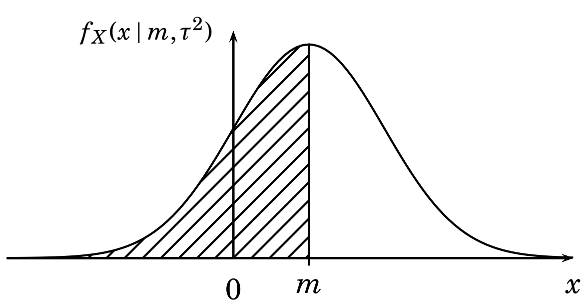
Question 3 – Let X be a discrete r.v. with distribution $P(X = x),\ x \in \mathbb{N}$.
Express the cumulative distribution function of X

$F_X(x) =$

Question 4 – Let X be a continuous r.v. with p.d.f. $f_X(x)$ defined on $\mathbb{R}$.
Express the cumulative distribution function of X

$F_X(x) =$

Question 5 – Let consider $X \sim \mathcal{N}(m, \tau^2)$ which p.d.f. is represented below.



1. What is the value of the hatched part on the graph?
2. What is the value of $P(X > m)$?

Question 6 – Let X be a r.v. such that $EX < +\infty$ and $Var(X) < +\infty$. Show that :

$$Var(X) < E[(X-a)^2] \text{ for all } a \neq EX.$$

Hint : write $VarX = E\left[(X-a+a-EX)^2\right]$.

Question 7 – Show that $f_X(x) = x^3$, $x \in [0, \sqrt{2}]$ is a probability density function.

Compute $E(X)$:

Compute $Var(X)$:

Question 8 – If X and Y have the joint density function

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{8\pi^2} \exp\left\{-\dfrac{x^2+y}{4\pi}\right\}, & \text{if } -\infty < x < +\infty, y \geqslant 0 \\ 0, & \text{otherwise.} \end{cases}$$

1. Find the density of $f_X(x)$ and give its name and parameters.

2. Find the density of $f_Y(y)$ and give its name and parameter.

3. Are X and Y independent?

Question 9 – Let $X_1,\ldots,X_n$ be a sample of n i.i.d. r.v.'s following a Poisson distribution with parameter 1.

1. Find the distribution of $S_n = \sum_{i=1}^{n} X_i$.

   Hint : $\big(X \sim \mathscr{P}(\lambda)\big) \Longleftrightarrow \big(M_X(t) = \exp\{\lambda(e^t - 1)\}$ is the moment generating function of $X\big)$.

2. Apply the CLT to obtain : $\lim_{n\to+\infty} P(S_n \leqslant n)$.

3. Deduce from 2. that $\lim_{n\to+\infty} \sum_{k=0}^{n} \frac{n^k}{k!}\, e^{-n} = \frac{1}{2}$.

Question 10 – Let $X_1,\ldots,X_n$ be n i.i.d. r.v's such that X_i, $i = 1,\ldots,n$, has a binomial distribution with parameter (N,p). We know that : $E(X_i) = Np$ and $Var(X_i) = Np(1-p)$. Let us consider the statistc $T_n = \dfrac{1}{(n-1)N}\sum_{i=1}^{n} X_i$, an estimator of p.

1. Show that T_n is an asymptotically unbiased estimator.

2. Show that T_n converge to p in probability.

Question 11 – Let $X_1,\ldots,X_n$ be n i.i.d. r.v's with the following distribution :

$$f_{X_i}(x) = \theta\, \frac{a^\theta}{x^{\theta+1}}\, \mathbb{1}_{[a,+\infty[}(x),\ a>0,\ \theta>1,\ i=1,\ldots,n,$$

where $\mathbb{1}_{[a,+\infty[}(x) = 1$ if $x \geqslant a$, 0 otherwise.

1. Compute the expectation of X_i : $E(X_i)$, $i=1,\ldots,n$.

2. Use the method of moments to propose an estimator for θ.

3. We want to find a maximum likelihood estimator for θ.

   (a) Write the likelihood.

   $L(\theta) =$

(b) Calculate the log-likelihood.

$$\log L(\theta) =$$

(c) Write the likelihood equation.

$$\frac{d}{d\theta} \log L(\theta) = 0 \Longleftrightarrow$$

(d) Solve the likelihood equation to express the MLE as $\widehat{\theta} = \frac{n}{\sum_{i=1}^{n} \log(x_i/a)}$.

Question 12 – Let us consider a statistical bayesian model. Let $f(x \mid \theta)$ be the distribution of the observation and $\pi(\theta)$, the prior distribution.
Express the posterior distribution : $\pi(\theta \mid x)$.

Question 13 – A soft-drink container filling machine is called "“under control"” if it fills on the average 12.2 ounces per container with standard deviation at most 0.05 ounces. A random sample of $n = 26$ containers yields $\overline{X}_n = 12.102$ and $\widehat{\sigma}_n^2 = 0.00425$.
Based on this information, and at level $\alpha = 0.05$, is the filling machine under control as to variability of fill?

Hint : For $\alpha = 0.05$, $\chi_{25}^{2^{-1}}(1-\alpha/2) = 40.64$, $\chi_{25}^{2^{-1}}(1-\alpha) = 37.65$, $\chi_{25}^{2^{-1}}(\alpha) = 14.61$, $\chi_{25}^{2^{-1}}(\alpha/2) = 13.12$.

Write the test of hypothesis :

H_0 : against H_a :

Compute the test statistics in this case.

Express the critical region.

Write a conclusion to answer the question.

Reminder

- **Poisson distribution**

$$\left(X \sim \mathscr{P}(\lambda)\right) \Longleftrightarrow \left(P(X=x) = \frac{\lambda^x}{x!}e^{-\lambda},\ x \in \mathbb{N}\right).$$

$E(X) = Var(X) = \lambda.$

- **Exponential distribution**

$$\left(X \sim \mathscr{E}(\theta)\right) \Longleftrightarrow \left(f_X(x) = \frac{1}{\theta}e^{-x/\theta},\ \theta > 0,\ x \in \mathbb{R}^+\right).$$

$E(X) = \theta,\ Var(X) = \theta^2.$

- **Gaussian Integral**

$$\int_{-\infty}^{+\infty} e^{-t^2/a}dt = \sqrt{a\pi},\ a > 0.$$

- **Chebyshev's Inequality** – Let X be a r.v. such that $EX < +\infty$ and $Var(X) < +\infty$. Then :

$$\forall \varepsilon > 0,\ P(|X - EX| > \varepsilon) = \frac{Var(X)}{\varepsilon^2}.$$

- **Convergence in probability** – The sequence do r.v. X_n converge in probability to a iff

$$\forall \varepsilon > 0, \lim_{n \to +\infty} P(|X_n - a| > \varepsilon) = 0.$$

- **Central Limit Theorem** – Let $X_1, \cdots, X_n$ be n independent and identically distributed (i.i.d.) r.v.'s with $E(X_1) = m$ and $Var(X_1) = s^2 > 0$ (both finite).
Then :

$$\sqrt{n}\left(\frac{\overline{X}_n - m}{s}\right) \sim \mathcal{N}(0,1) \text{ when } n \to +\infty.$$

- **Testing normal variance**

Testing variance
$H_0 : \sigma^2 = \sigma_0^2$

Test Statistics
$\Sigma_n = (n-1)\dfrac{\sigma_n^2}{\sigma_0^2}$

$H_1 : \sigma > \sigma_0$	$H_1 : \sigma < \sigma_0$	$H_1 : \sigma \neq \sigma_0$
Critical region $\Sigma_n > \chi^{2^{-1}}_{n-1}(1-\alpha)$	Critical region $\Sigma_n < \chi^{2^{-1}}_{n-1}(\alpha)$	Critical region $\Sigma_n < \chi^{2^{-1}}_{n-1}(\alpha/2)$ or $\Sigma_n > \chi^{2^{-1}}_{n-1}(1-\alpha/2)$